A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 27

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** Local people had used the area for fuelwood, timber for house building, clearings for farming, hunting and the collection of medicines and fibres, all of which protection could have stopped **1** Local people are employed as rangers, forestry workers and in ecotourism, so the forest is worth more to them standing than cleared **1** Ecotourist groups eat in local restaurants, are fed on local produce and buy locally made souvenirs, spreading the income through the community **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Trees are only felled where other trees of the same species stand upwind of the clearing, so their seed will blow in and recolonise it **1** Small trees are left standing so that they can grow and be harvested in the future when they are larger **1** Selected large trees are left because they produce many seeds that fill the gaps left by the trees that were felled **1** No mechanical or harvesting activity takes place near rivers, because aquatic ecosystems are ecologically sensitive **1** The use of large machines is restricted because they damage the trunks and roots of the trees that are not being felled **1** Track routes are planned to avoid important trees and their roots, including rare trees and those that feed monkeys or birds, and old tracks are re-used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	4
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** The tag allows the timber to be tracked from the stump where it grew to its final retail destination, such as a DIY store in the UK **1** That chain of custody is the evidence the FSC needs that the wood really came from a sustainably managed forest and was not mixed with illegally logged timber **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Ecotourism:** visitors pay to see the forest and its wildlife, and local businesses supply them **1** Timber confiscated from illegal loggers by the rangers is sold, and the money pays for conservation work **1** Electricity companies, airlines and individuals fund an area of forest as a carbon sequestration research project, because they want to offset atmospheric carbon dioxide **1** Lower-value species such as zericote are developed for furniture, musical instruments and carving, and small branches and offcuts are given to local artisans **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
01.5	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** The canopy is only opened in small gaps, so the forest microclimate - shade, humidity and temperature - is maintained and shade-dependent species survive **1** The soil keeps its cover of vegetation and leaf litter, and the roots that bind it, so erosion, leaching and sedimentation of rivers are far lower **1** Seed trees and small trees remain, so the gaps are recolonised by the same species and the forest regenerates naturally without replanting **1** Habitats and food sources are left in place and the forest is not fragmented, so populations of birds and mammals are not isolated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leaching** → nutrients or ions washed down out of the soil / washed away by water	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369, 386-387; AQA spec 3.5.3.3.	
02.1	**1 mark each** 1 mark for each of the 2 points below. **1 In-situ:** conserving species in their natural habitat with the whole community – ex-situ: away from the natural habitat, eg zoos, botanic gardens, seed banks, captive breeding **1** Needed when conservation in the wild is impossible or insufficient to prevent extinction (habitat gone, population too small, poaching) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **captive breeding** → breeding in captivity to increase numbers for release **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	2

Do not write outside the box

Question	Marking guidance	Marks
02.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Is the wild population threatened? **1** Is there a genetically diverse captive population? **1** Is in-situ conservation failing? **1** Is keeping a captive population realistic (habitat size, food, relationships, cost)? **1** Is release likely to be successful now or in future (suitable habitat, threats removed)? **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **captive breeding** → breeding in captivity to increase numbers for release **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	4
02.3	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Blue whale:** habitat size far too large to provide – food (huge quantities of krill / plankton) cannot be supplied – cost **2 Large blue butterfly:** complex inter-species relationship - larvae are taken into ant nests where they overwinter feeding on ant grubs – requires specific ant species and thyme food plant **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-34; AQA spec 3.1.2.3.3.	
03.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 2 400 / 2 000 **1** = 1.2 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • **food conversion ratio** → the feed needed per unit of food produced **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 3 points below. **1** Carnivorous species have a high market value, so they are the most profitable to farm, and consumers in wealthy countries prefer them **1** They are fed on fishmeal and fish oil made from wild-caught fish, so farming them still depends on catching wild fish and does not take the pressure off marine stocks **1** Several tonnes of wild fish may be needed per tonne of salmon, so the extra trophic level means the total amount of fish protein available falls rather than rises **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming • **trophic level** → the feeding level in a food chain **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
03.3	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Uneaten feed and faeces fall through the cages and accumulate on the sea bed, where their decomposition removes oxygen and smothers the benthic community **1** The nutrients released cause local eutrophication and can trigger algal blooms **1** Parasites, especially sea lice, build up in the dense stock and spread to wild salmon and sea trout passing the cages **1** Antibiotics, pesticides used to treat sea lice and antifouling chemicals from the nets disperse into the water and affect non-target species **1** Escaped farmed fish interbreed with wild populations, diluting the locally adapted gene pool, and compete with them for food and spawning sites **1** Predators such as seals and cormorants attracted to the cages are shot or trapped **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **eutrophication** → nutrient enrichment of water leading to algal growth • **gene pool** → the range of alleles / genes present in a population **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4

Do not write outside the box

Question	Marking guidance	Marks
03.4	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1 Integrated multi-trophic aquaculture:** seaweeds and filter feeders such as mussels are grown alongside the cages and take up the dissolved nutrients and particulate waste as their own food **1** Siting cages where the current is strong, and leaving sites fallow between cycles, so wastes are dispersed and the sea bed recovers **1** Lowering stocking density and vaccinating the fish, so less medication is needed, and using cleaner fish such as wrasse to remove sea lice instead of pesticides **1** Stronger, better maintained nets and predator exclusion nets, so fewer fish escape and predators do not have to be killed **1** Replacing part of the fishmeal in the feed with plant or insect protein, reducing the demand on wild fish stocks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
03.5	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Farming carnivorous species still depends on wild fish for feed, so expanding it increases rather than reduces the catch of wild fish **1** Suitable sheltered coastal sites are limited, and they compete with other uses and with sensitive habitats such as mangroves, which shrimp farming has already destroyed on a large scale **1** Some valuable species, such as tuna, cannot yet be bred reliably in captivity, so farms are stocked with wild-caught young fish and the wild population is still exploited **1** The local pollution, disease and escape problems mean that simply scaling the industry up multiplies its impacts **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-359; AQA spec 3.5.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 2 points below. **1** The scatter / variability of results around each mean – about 68% of values lie within one SD of the mean **1** The SDs are large relative to the difference in means (1.9 g) so the ranges overlap greatly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean	2
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** t-test **1** Comparing two means of measured (continuous) data **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **t-test** → a test for a difference between the means of two normally distributed sets **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** There is a 30% probability that the difference occurred by random chance **1** Above the 0.05 (5%) level so the difference is not significant – the null hypothesis (no difference) is accepted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **random** → positions chosen using random numbers so the choice is not biased • **null hypothesis** → a statement that there is no significant difference or correlation **Do not credit** • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	2

Do not write outside the box

Question	Marking guidance	Marks
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Two correlated variables may both be caused by a third variable that was not investigated **1** Significance only shows the result is unlikely to be chance – a mechanism / controlled experiment is needed for causation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **statistically significant** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **correlation** → a relationship in which the two variables change together • **causal** → one variable actually causing the change in the other **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test	2
04.5	**1 mark each** 1 mark for each of the 2 points below. **1** Increase the number of replicates (reduces the effect of variability) **1** Standardise other variables (soil, water, light) more closely to reduce scatter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
05.1	**1 mark each** 1 mark for each of the 4 points below. **1** Zooxanthellae are photosynthetic algae living inside polyp cells (up to 30 000 per mm^2) **1** They use the polyp's CO2 and excretory wastes (nitrogen, phosphorus) for photosynthesis and growth, so nutrients are recycled efficiently **1** They supply the polyp with glucose and oxygen - over 95% of its food energy – the polyp provides a protected habitat and nutrients **1** Polyps also trap plankton with cilia and stinging nematocysts – surplus CO2 is used to build the calcium carbonate skeleton **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** High light (shallow, clear water) for the algae – warm stable temperatures (about 18-30 °C) – low turbidity – constant salinity – well-oxygenated water **1** Found only where all these coincide (about 1% of ocean area) **1** Small changes (warming, sediment, freshwater runoff) move conditions outside the narrow range of tolerance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **range of tolerance** → the range of conditions in which a species can survive	3
05.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1 Fisheries:** support over a quarter of marine species including fish, lobsters, conch **1 Erosion protection:** complex structure dissipates wave energy protecting low coral islands – coral sand builds shores **1 Medicinal discoveries:** AZT and Ara-C from sponges, dolastatin from sea hares, bryostatin from Bugula, trabectedin from sea squirts, secosteroids, coral for bone grafts **1 Climate control:** carbonate skeletons sequester CO2 **1** Tourism (diving, snorkelling) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-60; AQA spec 3.1.2.3.4.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Environmental DNA (eDNA) analysis	1
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Transmitters are too heavy for small birds / their batteries would not last for a long-term study **1** Geolocator tags are small and only record time and light levels – latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Each separate (regional) population has its own unique gene pool **1** DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	
07.1	**1 mark each** 1 mark for each of the 4 points below. **1 SSSI:** UK legislation (Wildlife and Countryside Act) – best examples of UK flora, fauna, geological features – management plan agreed with owner who must notify Operations Likely to Damage (ploughing, drainage, pesticides, tree planting / removal) **1 SPA:** EU Birds Directive – places of international importance for birds (with SACs under the Habitats Directive form the Natura 2000 network; may restrict activities in surrounding areas) **1 Ramsar:** Ramsar Convention (intergovernmental agreement) – internationally important wetlands, originally for waterfowl, now wetland conservation and wise use generally **1 MCZ:** UK marine legislation (Marine and Coastal Access Act) under the W&CA framework – protects marine habitats / species by restricting activities such as fishing and dredging – NNR: best SSSIs with complete communities **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4
07.2	**1 mark each** 1 mark for each of the 3 points below. **1 Legal status:** restrictions on damaging activities within the area (and sometimes in surrounding areas, eg drainage) **1** Management agreements between owners and statutory bodies to maintain suitable conditions **1** But it does not protect against external influences such as air pollution or climate change – very few sites are state-owned so protection depends on owner cooperation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 3 points below. **1** Much of the UK's wildlife lives in habitats created by farming (hay meadows, hedgerows) which only survive if the farming continues **1** Managing land for wildlife costs money / reduces income so farmers need payments (points for beetle banks, hedgerows, buffer strips, wild bird seed, low-input grassland) **1** Benefits everyone so society pays through central funding – legislation cannot compel positive management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30, 38-39; AQA spec 3.1.2.3.2.	
08.1	**1 mark each** 1 mark for each of the 2 points below. **1 Moral belief:** protecting the environment is the 'right' thing to do regardless of benefit to people **1 Enlightened self-interest:** protecting the environment is done because it causes fewer difficulties (for people) than not doing so **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Total impact = population size x the individual (per capita) impact of each person **1** Per capita impact rises with affluence / access to energy, so both a larger population and a more affluent population increase the total impact **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
08.3	**Any 1 of 3** 1 mark each for any 1 of the 3 points below, in any order, maximum 1. **1** Slash and burn agriculture **1** Fishing **1** Discharge of sewage / biodegradable pollutants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **biodegradable** → broken down by decomposers **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370; AQA spec 3.6.1.	
09.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 C:** Waste products that are the raw materials for other processes	1
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) **1** The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO_2 **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Incineration uses high temperatures / fuel and releases pollutants – nutrients are not returned as usable raw materials **1** Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Succession:** pioneers, seres (hydrosere, lithosere, psammosere), modification of abiotic conditions, climax communities controlled by climate – deflected succession and plagioclimaxes • **Application:** deciding whether to manage (plagioclimax habitats need grazing, mowing, burning, coppicing, pollarding continued) or leave alone (climax habitats) – habitat creation planning the stages – controlling scrub / secondary succession – understanding that new habitats change over time • **Population dynamics:** birth and death rates, density-dependent and independent factors, carrying capacity, sigmoid growth, predator-prey cycles, r- and k-selection, MSY • **Application:** setting sustainable harvest levels (fisheries, whaling, forestry) – recognising k-selected species need strict protection and recover slowly – deciding when culling is needed (deer without wolves, introduced species) or when releases / captive breeding are needed (low breeding rate) – predicting effects of reintroducing predators • Monitoring population size, distribution, survival rate and age structure to assess success – adaptation and range of tolerance guiding abiotic management • **Evaluation:** uncertainties (natural fluctuations, unknown carrying capacity) – examples (Yellowstone wolves, Scottish deer, chalk grassland) – judgement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9

Do not write outside the box

Question	Marking guidance	Marks
	• **succession** → the gradual change in the community over time • **abiotic** → non-living / physical factors • **deflected succession** → a community held back from its climax by human activity **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	
	Model answer - Level 3 (8 marks)	
	Succession is the sequence by which pioneers colonise bare ground, modify the abiotic conditions and are replaced by later seres until a climax community controlled by climate is reached; where grazing, mowing, burning or coppicing halts it, a plagioclimax forms. This tells a conservation manager whether to intervene: chalk grassland, heathland and coppiced woodland are plagioclimaxes that disappear without continued management, whereas a climax habitat is best left alone, and habitat creation must plan the stages, control scrub and manage secondary succession.	
	Population dynamics - birth and death rates, density-dependent and independent factors, carrying capacity, sigmoid growth, predator-prey cycles and r- and k-selection - inform decisions of a different kind. Maximum sustainable yield sets harvest levels in fisheries, whaling and forestry; k-selected species that breed late and slowly need strict protection and recover slowly; culling may be justified where predators are absent, as with Scottish red deer without wolves, or where introduced species threaten natives, while the reintroduction of wolves to Yellowstone shows how restoring a predator changes vegetation through the food web. Monitoring population size, distribution, survival and age structure measures success, and knowledge of adaptation and ranges of tolerance guides how abiotic conditions are managed.	
	The limits are uncertainty: natural fluctuations, unknown carrying capacities and the difficulty of predicting how communities respond. Ecological understanding cannot remove that uncertainty, but without it conservation decisions - to manage or not, to cull or protect, how much to harvest - are guesses.	
	Weaker answer - Level 2 (5 marks)	
	Succession is when a habitat changes over time from pioneer species to a climax community. Conservationists need to understand this because some habitats like heathland will turn into woodland if they are not managed by grazing or burning. Population dynamics is about how populations grow and are limited by factors like food and predators. This helps conservation because you can work out how many animals can be taken without the population declining, and when a population needs protecting. Monitoring populations shows if conservation is working.	
	What makes the difference	
	• Level 2 explains both concepts correctly and links them to decisions, but generally; Level 3 uses the technical vocabulary (seres, plagioclimax, k-selection, MSY) with named cases (Yellowstone wolves, Scottish deer, chalk grassland).	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3 covers more applications (habitat creation, culling introduced species, adaptation and tolerance) and evaluates the uncertainties.	
	• Structure: Level 3 gives each concept its own paragraph with the decisions it informs, then a conclusion.	
10.2	1 mark for the point below. **1** Any taxonomic group of organisms classified together by similarity (eg species, genus, family) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-85; AQA spec 3.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how human activities influence biodiversity, using examples from a range of different contexts.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Direct exploitation:** food (fisheries, bushmeat), fashion, entertainment / pets, furniture and ornaments, traditional medicines, other products (whale oil) – deliberate eradication of predators, competitors, vectors, pests • **Changes in abiotic factors:** water availability (drainage, abstraction, HEP), light (forest clearance), oxygen (organic pollution, thermal), nutrients (eutrophication), pH (acid mine drainage, acid rain), temperature (thermal pollution, climate change), turbidity, physical damage – inactivity (stopping plagioclimax management) • **Changes in biotic factors:** introduced competitors, predators, pathogens, hybridisation – loss of pollinators, seed dispersers, keystone species – food chain impacts (sea otters, puffins, jellyfish) • **Habitat destruction:** deforestation, farmland expansion, urbanisation, mining, reservoirs – fragmentation and edge effects • **Contexts:** temperate woodland, rainforest, coral reefs, oceanic islands, mangroves, Antarctica	25
	– agriculture, fishing, forestry • **Positive influences:** habitat creation, conservation management, reintroductions, legislation • Evaluation of which influences are most damaging and hardest to reverse – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26, 328-330, 341-344, 367-369; AQA spec 3.1.2.2, 3.5.1.3, 3.5.2.2, 3.5.3.3.	

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Question	Marking guidance	Marks
11.2	**Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Definition of sustainability (Brundtland) – population x per capita impact – dynamic equilibria, feedback, tipping points, diversity • **Principles of natural systems:** renewable low-density energy, low temperatures, material cycles, non-toxic wastes – contrast with linear human systems • **Circular economy principles:** cycling materials, renewable energy, supporting ecosystems, separation of technical and biological materials, diversity, connected systems, design for extended use and re-use, optimum production, new technologies • Application to land use, water, minerals, waste, pollution, energy, food with examples (Kalundborg, Eigg, Ol Pejeta, Rio Bravo, EU initiatives, DC grids) • **Measuring progress:** ecological footprint, biocapacity, global hectares, Earth Overshoot Day, carbon footprint, Living Planet Index • **Barriers:** economics, competition, consumer behaviour, need for international agreement, existing infrastructure • Judgement on the extent achievable	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-389; AQA spec 3.6.	

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